

VLF reception, station geometry and hourly filtering

Notes for a SuperSID monitoring station

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1 Induced voltage in the loop antenna

A vertically polarized wave in the waveguide carries a horizontal magnetic field that is perpendicular to its direction of travel (Loudet, 2013). The voltage induced in the loop is:

$$\left[\frac{V_{\text{rms}}}{\text{V}} \right] = 2\pi \left[\frac{f}{\text{Hz}} \right] N \left[\frac{A_{\text{eff}}}{\text{m}^2} \right] \left[\frac{B}{\text{T}} \right] \cos \theta \quad (1)$$

where:

- V_{rms} is the root mean square voltage across the loop.
- f is the transmitter frequency.
- N is the number of turns in the loop.
- A_{eff} is the effective area enclosed by one turn.
- B is the magnetic flux density of the arriving wave.
- θ is the angle between the loop axis and the direction to the transmitter, so the response falls to zero when the loop faces it edge on.

The $\cos \theta$ term is why the bearing to each transmitter matters when the antenna is put up, and the two sections below are how that bearing and the path length are worked out.

2 Forward azimuth to a transmitter

$$\left[\frac{\theta}{\text{rad}} \right] = \arctan \left(\frac{\sin(\Delta\lambda)}{\cos(\phi_1) \tan(\phi_2) - \sin(\phi_1) \cos(\Delta\lambda)} \right) \quad (2)$$

where:

- θ is the forward azimuth in radians.
- ϕ_1 is the latitude of the starting point.
- ϕ_2 is the latitude of the destination point.
- $\Delta\lambda$ is the difference in longitude between the two points.

3 Great circle distance

$$\left[\frac{d}{\text{km}} \right] = \left[\frac{R_E}{\text{km}} \right] \arccos \left(\sin \phi_1 \sin \phi_2 + \cos \phi_1 \cos \phi_2 \cos(\lambda_2 - \lambda_1) \right) \quad (3)$$

where:

- d is the Great Circle distance.
- R_E is the mean radius of the Earth (≈ 6371 km).
- ϕ_1, ϕ_2 are the latitudes of the two points.
- λ_1, λ_2 are the longitudes of the two points.

Equations 2 and 3 are what `vlf_transit.py` evaluates for every transmitter in its table, which is how the installer can offer the ten or twenty nearest ones and how `vlf_table.pdf` lists a bearing and a distance for each.

4 Why the hourly figure says BEMA was applied here

The plot label reads **BEMA wing 6 (applied here, the csv itself is raw)** because the smoothing seen in the figure is performed by the plotting script, not by SuperSID. Although `log_type` may be set to `filtered` in `supersid.cfg`, that setting governs only the files written at the end of each UTC day. The hourly dump produced by `hourly_save` is written with `log_type='raw'` hardcoded in `supersid.py`, so the values in `hourly_current_buffers.*.csv` are exactly as captured from the sound card, and the file's own header confirms this with the line `# LogType = raw`.

To make the hourly figure comparable with the finished daily record, the script reproduces SuperSID's BEMA filter (`SidFile.filter_buffer`) on the data before drawing: for each sample it takes the lowest value within a window of $2 \times \text{bema_wing}$ points, then applies a single moving average of $2 \times \text{bema_wing} + 1$ points over those minima. With the default wing of 6 this spans roughly one minute at a five second log interval. Taking the minimum first causes the curve to follow the quiet background level, so a sudden ionospheric disturbance stands out as a departure from that baseline rather than being averaged away. Where a file already carries `LogType = filtered`, the script detects this from the header and applies nothing further, and the label changes accordingly.

Shorter version, for use as a figure caption

Signal from the hourly buffer, which SuperSID writes unfiltered regardless of the `log_type` setting. BEMA (wing 6: rolling minimum over 12 samples, then a 13 sample moving average) is applied by the plotting script to match the filtering used in the daily files.

References

Loudet, L. (2013). *SID monitoring station*. <https://sidstation.loudet.org/home-en.xhtml>